

Pose-Aware Self-Supervised Learning with Viewpoint Trajectory Regularization

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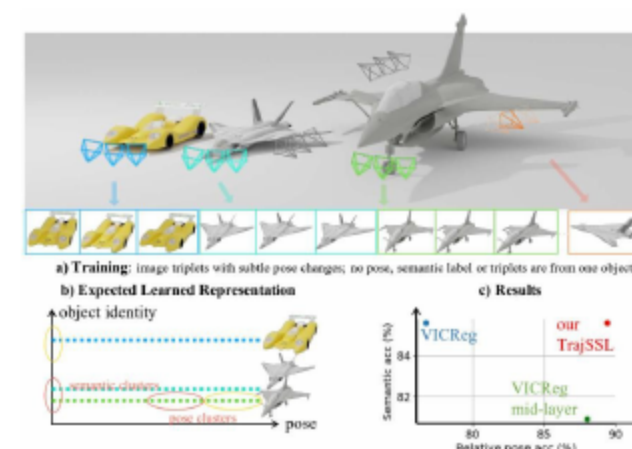
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Why Pose-Aware SSL: Contributions

3 SSL Overview and Problem Framing

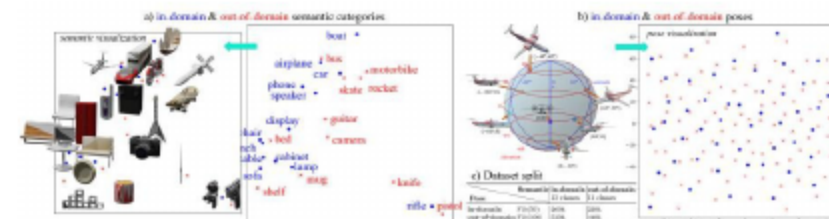
- SSL learns from unlabeled images for robust semantics
- Standard SSL maps varied views to one invariant feature
- But recognition also depends on viewpoint sensitivity
 - Evaluate representations for both class and pose alignment
 - We extend SSL to learn pose-aware representations
- Goal: keep category structure while encoding viewpoint

4 Method and Dataset at a Glance



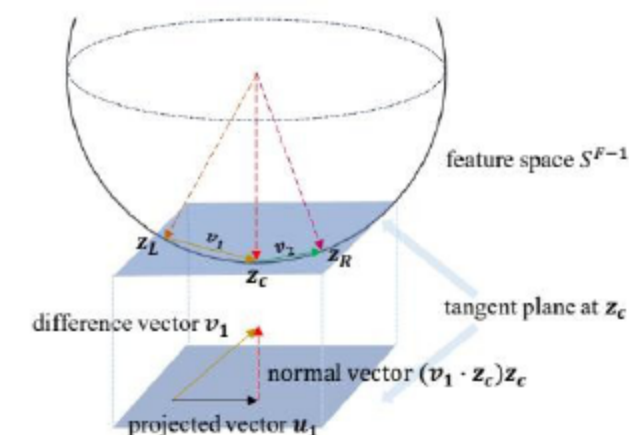
- Object-centric dataset of adjacent view triplets
 - No semantic or pose labels; mirrors natural seeing
 - Train on triplets; add viewpoint trajectory loss
- Encourages smooth feature-space motion along views
 - Preserves semantics while modeling relative pose
 - Superior generalization to novel objects vs SSL

- SSL splits into contrastive and non-contrastive camps.
- Most SSL targets semantics; geometry is underexplored.
- Geometry-aware SSL uses equivariance via autoencoders.
 - Prior pose methods depend on predictors or supervision.
 - We exclude GT labels; learn pose via trajectory constraints.
- Benchmark spans abs/rel pose on synthetic, OOD data.



- SSL trained without labels; evaluate geo+semantic
 - In-domain vs out-of-domain categories, clear splits
 - Semantic retrieval and pose accuracy evaluated
- Absolute and relative pose tasks assess generality
 - Pose alignment removes ambiguity in absolute evals
 - Relative pose metrics handle out-of-domain data

- Probe intermediate conv features to test pose cues.
- Mid-level embeddings align with geometric estimation.
- conv3 representations outperform high-level semantics.
 - Local embeddings capture pose while preserving semantics.
 - Dimensionality reduced via compression with negligible accuracy loss.
- Probes compare pose sensitivity and semantic consistency.

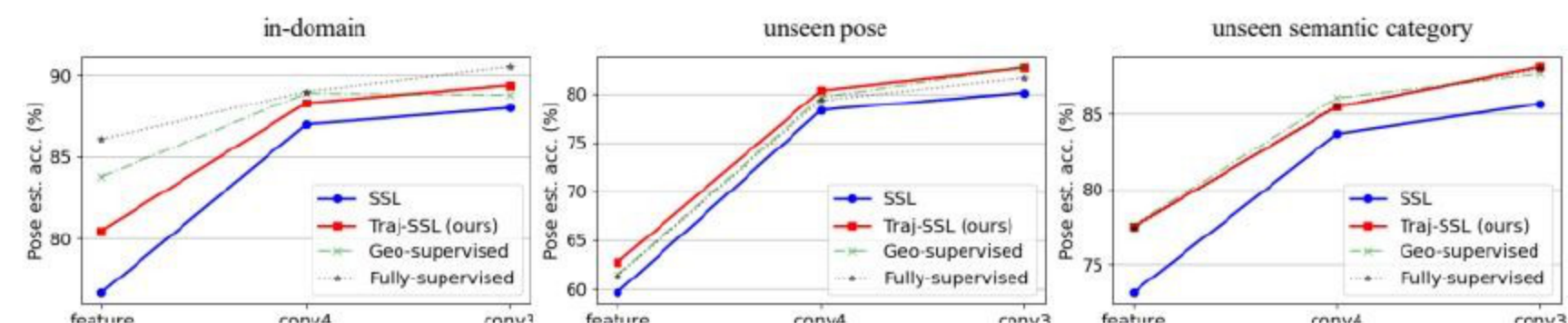


- Enforce smooth viewpoint trajectories in features
 - Geodesic alignment on the hypersphere manifold
 - Triplets align feature changes with pose changes
- Project difference onto tangent-plane directions
 - Maximize cosine similarity of projected steps
 - Combine unsupervised semantic and geo-trajectory losses

9 Training & Evaluation Protocols

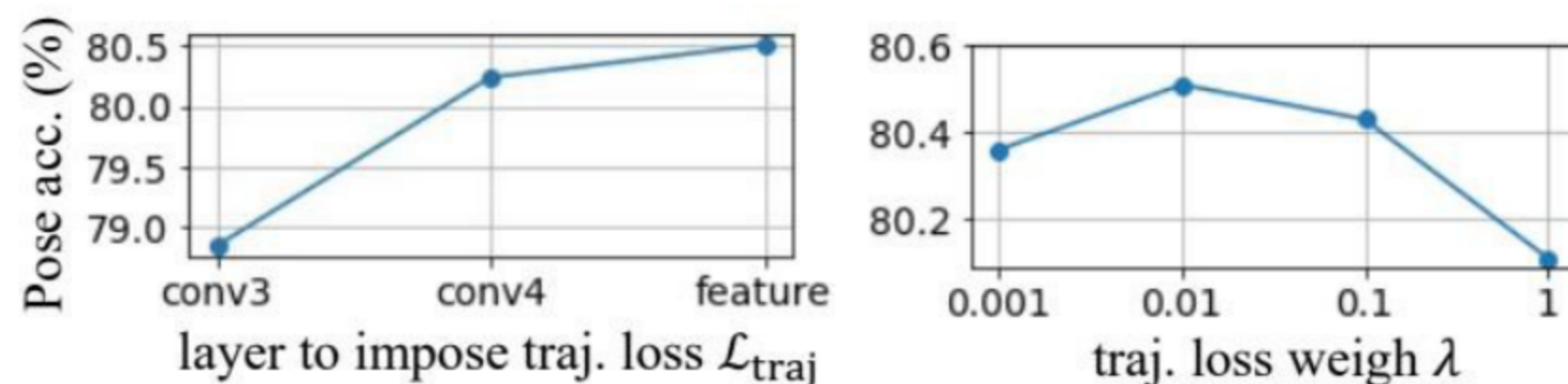
- Two-stage SSL; baselines: supervised, geometry, invariant SSL
- Trajectory-regularized pretraining on adjacent triplets
- Shared backbone and projector; combined SSL and pose losses
 - Semantic evaluation via linear, perceptron, and k-NN probes
 - Pose accuracy with absolute and relative estimation metrics
- Report across layers; OOD gains; in-domain, unseen pose/class

3 Results Across Settings and Layers



- Mid layers, esp. conv3, boost pose vs last features.
 - TrajSSL lifts pose and stays competitive semantically.
 - Out-of-domain gains are largest with mid layers.
- Mid layers close the gap to supervised baselines.

3 Ablations: Layer & Loss Weight



- Conv3 placement of trajectory loss gives best pose accuracy.
 - Mid-layer features close the OOD gap to supervised baselines.
 - Semantic classification stays stable across layer placements.
- Loss weight matters; balanced settings yield steady gains.

12 Visualizations and Key Takeaways



- Embeddings cluster semantically with pose ordering
 - Trajectory cues sharpen pose accuracy over SSL baselines
 - Category structure preserved; no semantic collapse
- Mid-layer trajectory loss yields gains across settings
 - Synthetic-trained models transfer to real 3D pose tasks
 - Nearest neighbors become more pose-consistent